Department of Electrical & Electronic Engineering

Rajshahi University of Engineering & Technology

# **LAB Report**

Course Code : **EEE 3100**

Course Title : **Electronics Shop Practice**

Experiment No. : 02

Experiment Name : PCB Design and Soldering a Charger Circuit

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**Experiment No.** 02

**Experiment Title:** PCB Design & Soldering a Charger Circuit

**Objectives:**

- To study the operation and observe the output of a charger circuit.
- To design a charger circuit on PCB board and to generate a required voltage as output.

**Theory:**

A cell phone charger is an essential electronic device designed to convert standard AC mains power (typically 220-240V in many regions) into a low-voltage DC power suitable for charging the battery of a mobile phone. This conversion process is accomplished through several key stages:

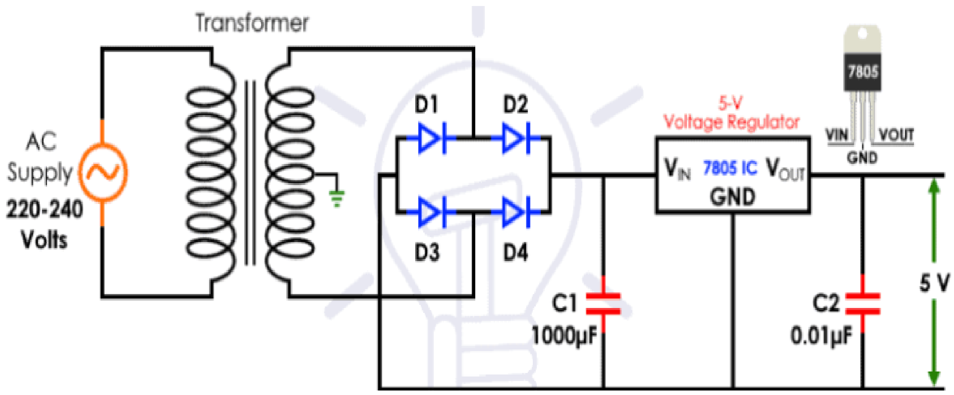


**Figure 2.1:** Charger Circuit

1. **Step-Down Transformation: The Role of the Center-Tapped Transformer**
   - The initial stage involves a transformer to reduce the high AC voltage to a lower, more manageable level. In this circuit, a center-tapped step-down transformer is employed.
   - A transformer operates on the principle of electromagnetic induction. It comprises two coils, the primary and the secondary, wound around a common iron core. The alternating current in the primary winding creates a changing magnetic flux, which, in turn, induces an alternating voltage in the secondary winding.
   - **Center-Tapped Transformer:** A center-tapped transformer has a crucial difference in its secondary winding. A center tap is a connection made at the exact midpoint of the secondary winding. This tap effectively divides the secondary winding into two equal halves.
     – In this specific case, a ”9-0-9” transformer is used. This designation indicates that the voltage between either end of the secondary winding and the center tap is 9V AC. The voltage across the entire secondary winding is 18V AC.

- The center tap serves as a reference point (often considered the ”ground” or ”neutral”) and plays a vital role in full-wave rectification.
- The key advantage of a center-tapped transformer is that it allows for full-wave rectification using only two diodes, as opposed to the four diodes required in a bridge rectifier configuration. This can sometimes simplify the circuit design.
- The center tap provides two output voltages that are equal in magnitude (12V in this case) but 180 degrees out of phase with each other.

2. **Rectification: Converting AC to Pulsating DC**

- The output of the transformer is AC, but mobile phone batteries require DC power. A rectifier circuit is used to convert the AC voltage into DC voltage.
- **Full-Wave Bridge Rectifier:** A full-wave bridge rectifier is commonly used. It consists of four diodes arranged in a bridge configuration.
  - During the positive half-cycle of the AC input, two diodes conduct, allowing current to flow in one direction through the load.
  - During the negative half-cycle, the other two diodes conduct, and current flows through the load in the *same* direction.
  - This process ensures that current flows through the load during both halves of the AC cycle, resulting in a more efficient conversion than half-wave rectification.
- The output of the rectifier is a pulsating DC voltage. This voltage is not smooth DC but rather a series of pulses that rise and fall with the AC input frequency.
- The pulsating DC voltage contains an AC component, known as ripple.

3. **Filtering: Smoothing the Pulsating DC**

- The pulsating DC voltage from the rectifier is not directly suitable for charging a mobile phone battery. A filter circuit is used to smooth out these pulsations and reduce the ripple, bringing the voltage closer to pure DC.
- A capacitor is the primary component in a simple filter circuit, and it is connected in parallel with the load.
- **Capacitor Action:**
  - During the rising portion of the pulsating DC voltage, the capacitor charges, storing electrical energy.
  - During the falling portion, the capacitor discharges, releasing the stored energy. This charging and discharging action smooths out the voltage variations, reducing the ripple.
- The effectiveness of the filter is directly related to the capacitance value. A larger capacitance results in a lower ripple voltage and a smoother DC output.
- The relationship between capacitance (C), current (I), time (t), and voltage (V) in a capacitor filter can be approximated by:

$$C = \frac{I \cdot t}{V} \tag{1}$$

Where:

- I is the load current (the current drawn by the mobile phone during charging).
- t is the time between peaks of the rectified waveform. This time is inversely proportional to the frequency of the AC supply (e.g., for a 50Hz supply, t would be related to 1/50 seconds).
- V is the peak-to-peak ripple voltage (the difference between the maximum and minimum voltage values in the filtered DC).

4. **Voltage Regulation: Maintaining a Stable Output**

- Even after filtering, the DC voltage may still exhibit slight variations due to fluctuations in the AC mains voltage or changes in the load current (as the phone battery charges). A voltage regulator is used to ensure a constant and stable output voltage.
- A linear voltage regulator, such as the 7805 integrated circuit (IC), is commonly employed in cell phone chargers.
- **7805 Voltage Regulator:**
  - The 7805 is a three-terminal device. The three terminals are typically labeled as Input, Output, and Ground (or Common).
  - It is designed to provide a fixed output voltage of 5V DC.
  - It has a specified input voltage range (e.g., typically between 7V and 35V). The input voltage must be higher than the desired output voltage for the regulator to function correctly.
  - It also has a maximum current rating (e.g., 1A). This is the maximum current that the regulator can supply to the load without overheating or being damaged.
- **Regulation Mechanism:** The regulator maintains a constant output voltage by dissipating any excess voltage as heat. For example, if the input voltage is 9V and the output voltage is 5V, the regulator dissipates the 4V difference.
- **Heat Dissipation:** Because the regulator dissipates power as heat ($P = V \cdot I$, where V is the voltage difference and I is the current), it can become hot, especially at higher input voltages or load currents.
- A heatsink is often required to dissipate this heat effectively. The heatsink is a metal component that is attached to the regulator to increase its surface area, allowing heat to be transferred to the surrounding air more efficiently. This prevents the regulator from overheating and ensures its reliable operation.

The cell phone charger circuit efficiently converts high-voltage AC power from the mains supply into a stable, low-voltage DC output suitable for charging mobile phone batteries. The process involves: using a step-down transformer to reduce the AC voltage; employing a full-wave bridge rectifier with four diodes to convert the AC voltage into pulsating DC; filtering the pulsating DC with a capacitor to reduce ripple; and utilizing a voltage regulator (like the 7805) to maintain a constant 5V DC output, even with variations in input voltage or load current.

**Required Apparatus:**

**Table 2.1:** Table For Required Apparatus

| SI No. | Component Name | Specification | Quantity |
|---|---|---|---|
| 1 | Diodes | 1N4007 (1000V PIV, 1A) | 4 |
| 2 | Capacitors | 100µF, Voltage Rating: 50V, Polarized | 2 |
| 3 | Integrated Circuit (IC) | 7805 (5V Output, 1A) | 1 |
| 4 | Transformer | Center-tapped transformer, 220-240V AC (50Hz) to 9-0-9 V | 1 |
| 5 | PCB | Printed charger circuit | 1 |
| 6 | Jumper Wires | - | As per need |
| 7 | Multimeter | - | 1 |
| 8 | Soldering Iron | - | 1 |

**Circuit Diagram:**

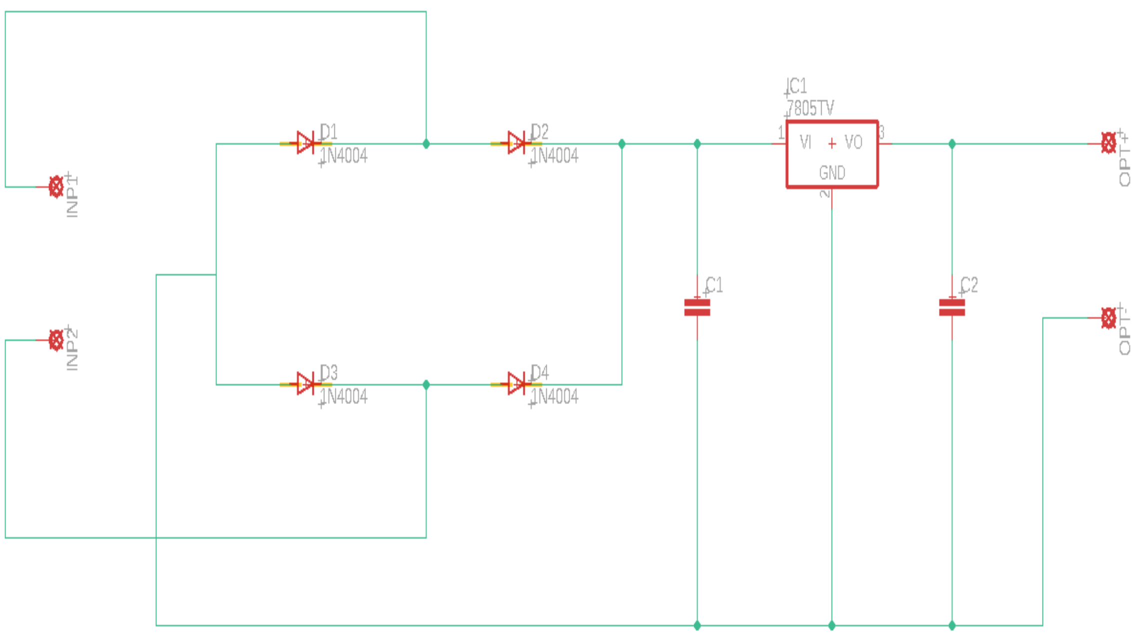


**Figure 2.2:** Circuit Diagram of Charger Circuit

**Experimental Setup:**

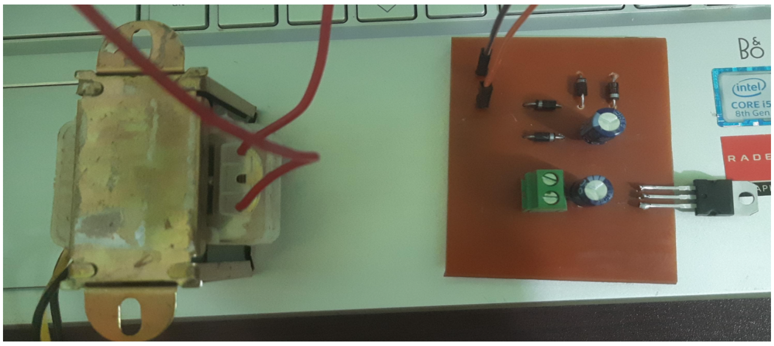

**Figure 2.3:** PCB Setup

**PCB Design:**

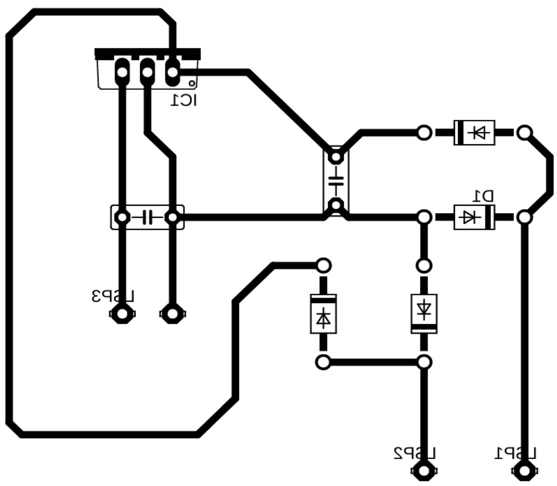

**Figure 2.4:** PCB Design

**Experimental Procedure:**

1. Designed the charger circuit on a PCB.
2. Soldered all the required apparatuses on the PCB.
3. Ensured all safety precautions are followed.
4. AC power was supplied though the center-tapped transformer.
5. Measured the AC voltage at the transformer's secondary terminals.
6. Measure the DC voltage and observe the output at the output terminals.

**Theoretical Result:**

1. Transformer Output Voltage ($V_{AC}$):
   - $V_{AC(peak)} = V_{AC(RMS)} \times \sqrt{2} \approx 9V \times 1.414 \approx 12.73V$
2. Rectifier Output Voltage ($V_{DC}$):
   - $V_{DC(peak)} = V_{AC(peak)} - 2 \times V_{diode} \approx 12.73V - 2 \times 0.7V \approx 11.33V$
3. Filtered Output Voltage:
   - $V_{filtered} \approx 11.33V$
4. 7805 Voltage Regulator Output:
   - $V_{output} = 5V$

**Experimental Result:**

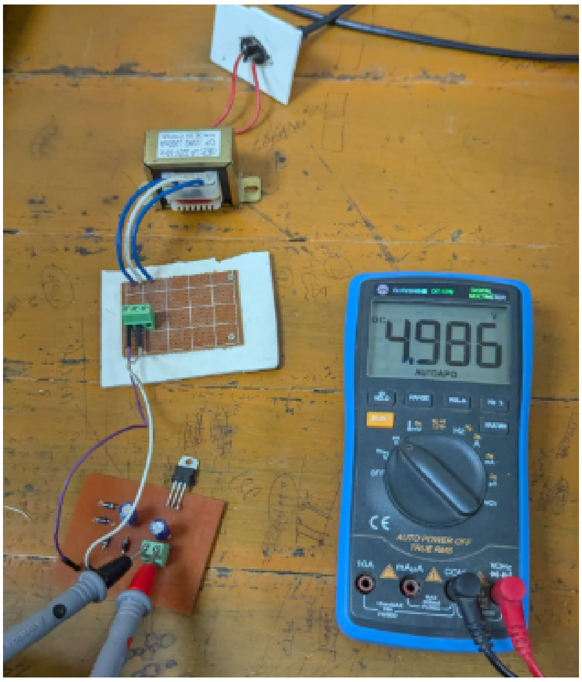


**Figure 2.5:** Experimental Result

**Discussion:**

In this lab, crafting a charger circuit by using PCB was the main objective. Unlike the previous experiment of astable multivibrator, good soldering was the top priority for this experiment. However, at first, the expected output (5V) wasn't achieved. After a few attempts of troubleshooting, it was found that the components, namely a diode and the IC 7805 weren't soldered according to their required polarity. Desoldering the components, they were fitted properly. Finally, the experiment was a success and had a output of almost 5V (precisely 4.986V).

**Conclusion:**

The charger circuit is contructed to achieve a certain magnitude of DC voltage after convertion of the AC power supply. The lab experiment discussed here helps to convert 220-240V AC (50Hz) to a magnitude of 5V DC. Using a center-tapped transformer, at first, the regular household AC power supply is stepped down to 12V AC (RMS). This voltage is then rectified with the help of full bridge rectifier using four diodes' cofiguration. The pulsating DC voltage, resulting from the rectifier, is smoothed by capacitor. Later on, this voltage is provided to IC 7805 to regulate it to 5V DC as the required output.

In the lab experiment, this whole process was executed successfully by a designed PCB where all of the necessary components were soldered and inter-connected.

As such, understanding the gist of the charger circuit's contruction, we can make our own cell-phone charger without any trouble.

**References**

[1] https://www.electricaltechnology.org/2021/02/cell-phone-charger-circuit.html